DEPARTMENT OF COMPUTER & SOFTWARE ENGINEERING

COLLEGE OF E&ME, NUST, RAWALPINDI

EC 312 – Digital Image Processing

Assignment # 01: Connected Component Labeling based Segmentation

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Lab Assessment

In-Lab	**Post Lab**			**Total**
	Data Presentation	**Data Analysis**	**Writing Style**	

Table of Contents

Task 1
Solution 1
Explanation of the Approach: 1
V-set Definition and Decision Process 1
Steps Followed for All Tasks 1
Flow Diagram 2
Results & Comparisons: 2
Sample Test Images 2
Experimental Results Table 3
Conclusion: 3

Task

1. **Define a V set. In first phase, Separate Optic Disc (OD) from the background using 8-connectivity based Connected Component Analysis (CCA). For this intelligently design V set using images from train folder and see how good they are working. This should give you 2 labels i.e. one for background and the other for complete OD.**

2. **Differentiate OD components. Define new groups of V sets again from train data to further divide the OD into Optic Cup and Optic Disc. Now you should have 3 labels in total, i.e. background, Optic Cup and Optic Disc**

3. **Compute dice coefficient. Once you are done with designing complete pipeline, use data from test folder to compute Dice Coefficient for each class i.e. background, Optic Cup and Optic Disc.**

Solution

Explanation of the Approach:

V-set Definition and Decision Process

The selection of the V-set (the set of intensity values belonging to the object of interest) was the most critical step in this segmentation process.

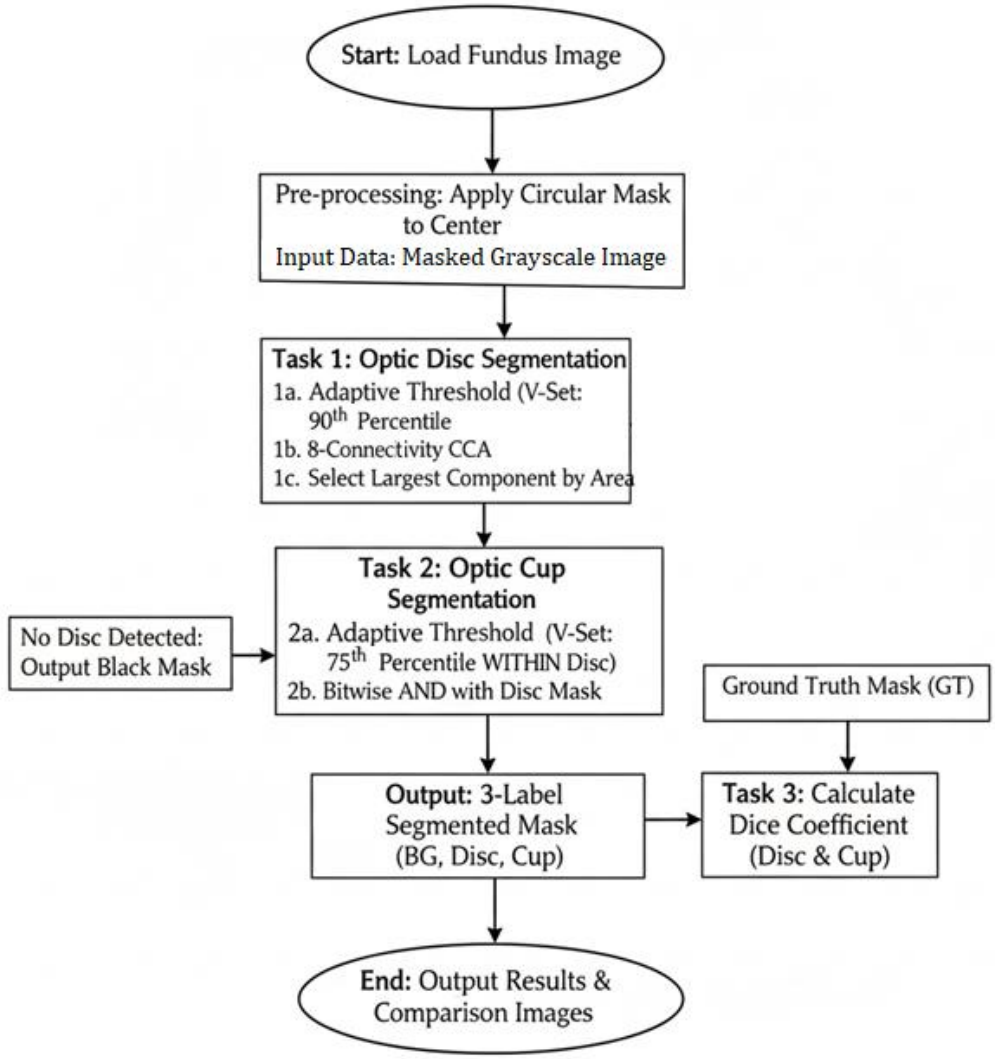
- **For Optic Disc (OD) Segmentation:** I defined the V-set based on the top 10% of pixel intensities (90th percentile) within the retina area. I chose this adaptive percentile approach because fixed intensity thresholds (such as $V > 200$) failed on images with poor illumination or lower contrast. Since the Optic Disc is anatomically the brightest region in the fundus, the 90th percentile consistently captures the Disc regardless of the overall brightness of the image.
- **For Optic Cup Segmentation:** I defined a secondary, more restrictive V-set. This was set to the top 25% of intensities (75th percentile) found exclusively within the region already identified as the Optic Disc. This internal V-set logic ensures that the algorithm only looks for the "brightest core" of the disc to identify the cup, preventing background noise from being mislabelled.

Steps Followed for All Tasks

1. **Masking and Pre-processing:** I applied a circular spatial mask to the centre of the fundus images. This was done to exclude the bright, white crescent-shaped artefacts often found at the edges of the camera view, which would otherwise skew the V-set selection.
2. **Optic Disc Extraction (Task 1):** I applied the 90th percentile threshold to create a binary image. I then used **8-connectivity Connected Component Analysis (CCA)** to label all objects. I programmed the logic to select only the largest connected component by area, which effectively isolated the Optic Disc and removed small noise fragments.
3. **Optic Cup Extraction (Task 2):** Within the area of the detected Optic Disc, I applied the 75th percentile V-set. This created a nested segmentation within the Disc boundaries.

4. **Evaluation (Task 3):** I automated the calculation of the Dice Coefficient by comparing my predicted masks against the expert ground truth masks provided in the dataset.

Flow Diagram



Flow Diagram

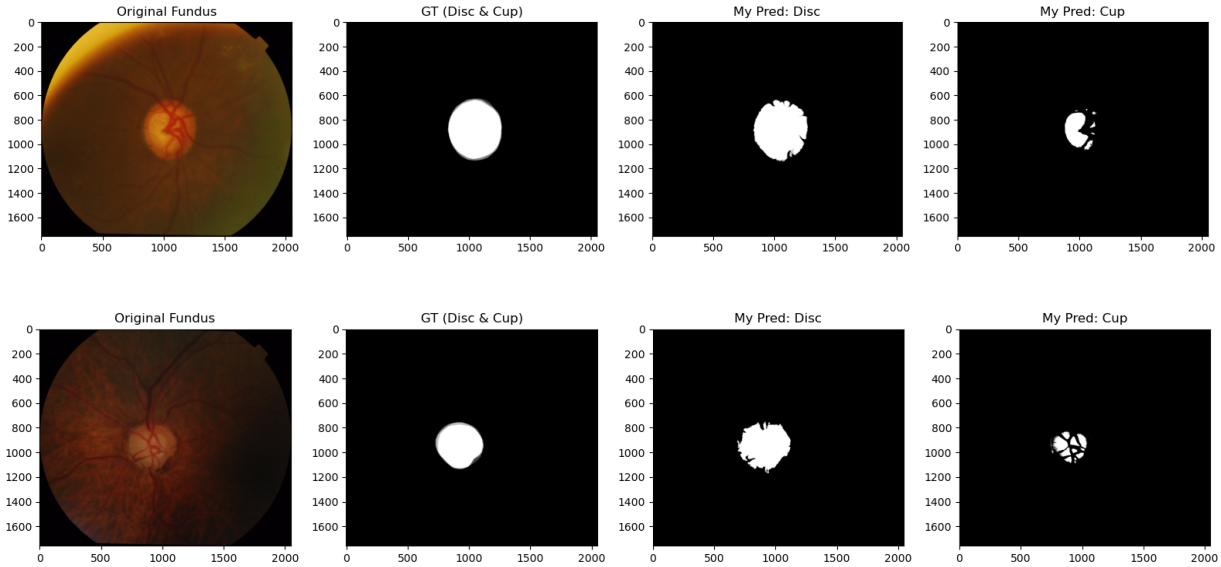
Results & Comparisons:

Sample Test Images

```
Final Results Table (Task 03)

                Experiment  Disc Dice  Disc Acc  Cup Dice  Cup Acc
   Experiment 1: Strict       0.2649    0.9751    0.0255   0.9797
 Experiment 2: Moderate       0.6846    0.9885    0.1736   0.9824
Experiment 3: Optimized       0.8790    0.9893    0.4117   0.9863
```

Console Output



Output Comparison Images

(The remainder of the output images is in the zip file submitted.)

Experimental Results Table

The following table summarizes the performance of the algorithm across different V-set configurations.

Experiment Name	Disc Percentile	Cup Percentile	Average Disc Dice	Average Cup Dice
1. Strict	98	95	0.2649	0.0255
2. Moderate	95	85	0.6846	0.1736
3. Optimized	90	75	0.8790	0.4117

Conclusion:

The results indicate that a 90th percentile V-set for the optic disc provides the most accurate balance for region extraction. Higher percentiles were too restrictive and failed to capture the full anatomy. The 8-connectivity CCA was essential for bridging gaps created by blood vessel shadows within the disc region. While the cup dice is lower than the disc dice, this is expected due to the low contrast between the cup and disc boundaries in many fundus images.